



Hall A BCM Linearity Procedure

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Estimated Time to Perform: ~60 minutes

Procedure Overview

This procedure is to be used by MCC operators to verify the linearity of the Hall A BCMs with respect to Faraday Cup #2 (5 MeV region) across a range of beam currents. This procedure is invasive; i.e., it will interrupt beam delivery to the other halls. Beam is first centered on FC#2 and data is recorded at 60-second intervals for descending test currents with FC#2 in. Beam is then sent to Hall A and data is taken at the same descending test currents for 90-second intervals, beginning with the maximum possible test current (as determined by the Hall Run Coordinator and the Program Deputy) and ending with 0.2 μA beam (refer to [Figure 2-1 on page 4](#) and [Figure 3-1 on page 5](#) for current vs. time plots of the procedure). The data taken will be analyzed “off-line” by Hall A personnel. The setup steps in Section 1.0 are non-invasive and can be performed before turning off beam to the halls.

This procedure is divided into sections as follows:

- Section 1.0 [Setup Steps \(non-invasive\) on page 2](#)
- Section 2.0 [Cross Calibrating the 0L02 BCM with FC#2 \(invasive\) on page 2](#)
- Section 3.0 [Datataking for the Hall A BCM Linearity Test on page 4](#)

Prerequisites

1. The request for the Hall A BCM linearity test must come from the Hall A Run Coordinator and be approved by the Program Deputy. Since this procedure interrupts beam delivery, the Hall A Run Coordinator or Program Deputy must first get approval from the other halls that are receiving beam.
2. 1–120 μA CW beam to Hall A (the maximum current level to be used for this test should be specified by the Hall A Run Coordinator).
3. CW beam delivery to Hall A has been set up according to the criteria established in the [Optics Restoration and Finalization Procedure](#) (i.e., you have clean CW transmission to Hall A).



Procedure Steps

1.0 Setup Steps (non-invasive)

1. Open the Current Monitoring in the Injector/Polarized Source screen (monticello.edl⇒Injector⇒INJ Current Monitoring) and connect FC#2 to its picoammeter by clicking on the Picoammeter button.
2. Start a Striptool to look at the signal from the FC#2 picoammeter (FCupsCORRECTED.VAL) and the 0L02 BCM (IBC0L02Current). Set the scale to look at the range within 300 nA of the maximum current specified for the test. This striptool is used later in the procedure and will provide the resolution necessary to see the changes in these signals.
3. Achieve the following conditions in Hall A:
 - Stop beam delivery to Hall A.
 - Mask the Hall A target-movement FSD bit as required for target removal.
 - Call Hall A and ask them to remove the target from the beam path (or verify that it is already OUT).
 - Mask the Hall A fast raster FSD bit on FSD node 1H002.
 - Turn OFF the Hall A fast raster.


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2.0 Cross Calibrating the 0L02 BCM with FC#2 (invasive)

CAUTION: The Hall A target must be removed from the beam path when performing this procedure.

1. Verify that FC#2 is in the “Picoammeter” mode (see Step 1, above). **THIS INSURES THAT THE PICOAMMETER SIGNAL IS AVAILABLE ACROSS THE FIREWALL SO THAT HALL A CAN ACQUIRE DATA.**
2. Call Hall A and verify that all necessary datataking signals are available to them.
3. Close the Hall B and Hall C slits.
4. Switch to CW beam delivery at the highest possible current (this should be specified by the Program Deputy in consultation with the Hall A Run Coordinator and will be $\geq 30 \mu\text{A}$).
5. Open the magnet screen for **MBH0L02H** and **V** (monticello.adl⇒Magnets⇒Magnet Commander⇒0L) and perform a Twiddle save of the present values for these magnets (these values will be referred to as the “FC#2_out” values in the remainder of this procedure). Also record these values in [Table 2-1 on page 3](#).
6. Insert FC#2.
7. Center the beam on FC#2 (**it should still be in picoammeter mode**) as follows:
 - a. Elog the present current on FC#2 (picoammeter mode) and the present Bdl settings of **MBH0L02H** and **V**.
 - b. Referring to the striptool of the FC#2 picoammeter and the 0L02 BCM that was started earlier ([Step 2 on page 2](#) in Section 1.0), adjust **MBH0L02H** to

maximize the current on FC#2 with respect to the current on 0L02 (see note below).

NOTE: The FC#2 and 0L02 traces on the striptool will both show ~1% of peak-to-peak noise (the average values can easily be estimated). Both traces may slowly drift as the actual beam current from the gun fluctuates (in the ~1% range). Since the ultimate goal is to optimize the beam position on FC#2, when adjusting MBH0L02H and V, it is necessary to eliminate the effect of the current fluctuations from the gun by directly comparing the readings (traces) from FC#2 and 0L02.

- c. Repeat Step b, above, for **MBH0L02V**.
- d. Adjust **MBH0L02H** in both directions to find the two points (one in the positive adjustment direction and the other in the negative) where the FC#2 trace dips 0.5 μA with respect to the 0L02 trace. Set the **MBH0L02H** Bdl to the average of the two points. **EXAMPLE:** If the two points are -142 and -172, then the average is -157.
- e. Repeat Step d, above, for **MBH0L02V**.
- f. If the FC#2 trace is below the 0L02 trace, is the difference <50 nA?



A. Repeat Steps b–f, above, to verify that **MBH0L02H** and **V** are optimized. If the FC#2 trace is still more than 50 nA below the 0L02 trace, abort the test and email Arun Saha (saha@jlab.org).

- g. Elog the present current on FC#2 and the 0L02 BCM and the new Bdl settings for **MBH0L02H** and **V**. Also record the present Bdl settings for **MBH0L02H** and **V** in Table 2-1, below (these values will be referred to as the “FC#2_in” values” in the remainder of this procedure).

Table 2-1: FC#2_out and FC#2_in Corrector Values

	MBH0L02H	MBH0L02V
FC#2_out		
FC#2_in		

8. Select appropriate test values between the maximum current and 0.2 μA (refer to Table 2-2 for some examples of appropriate test values). Generally speaking, the next test value is roughly half of the preceding test value, down to 0.2 μA .

Table 2-2: Example Test Values

Max. Current	Test Values (in μA)
40 μA	40, 20, 10, 5, 2, 1, 0.5, 0.2
100 μA	100, 50, 20, 10, 5, 2, 1, 0.5, 0.2
150 μA	150, 80, 40, 20, 10, 5, 2, 1, 0.5, 0.2

9. Using beam to FC#2, determine the appropriate attenuator setting for each test value and record them for later use (this eliminates delays during the “timed” steps later in this procedure).
10. Set the beam current to the maximum test value (FC#2 should still be inserted).



11. Insert FC#1 (this keeps beam from reaching IBC0L02).
12. Call the Hall A counting house and ask them to begin their data-logging process. After they inform you that they are logging data, wait 60 seconds and proceed to the following step (refer to Figure 2-1, below, for a graphical depiction of the entire data-logging process).

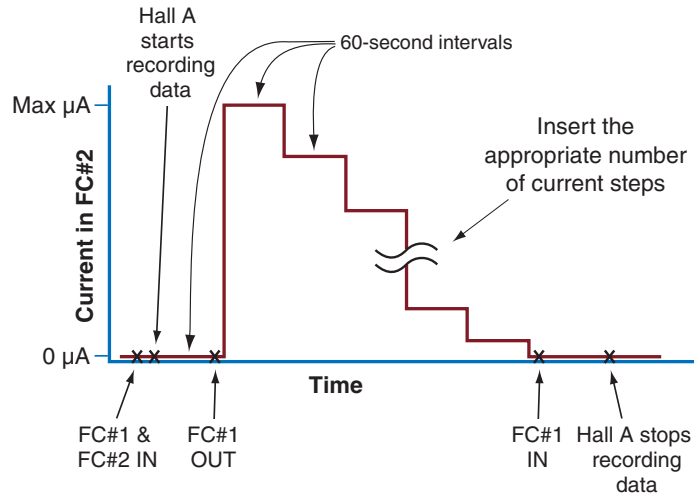


Figure 2-1: Graphical Depiction of the Data-Logging Process

13. Pull FC#1 out (leave FC#2 in), wait 60 seconds and then proceed to the following step.
14. Step the beam current to the next lowest test current and leave it there for 60 seconds. Repeat this for the remaining test currents that were determined in Step 8; after all test currents are completed, proceed to Step 15, below).
15. Insert FC#1 for 60 seconds.
16. Call the Hall A counting house and tell them to stop their data acquisition process.
17. Go to Section 3.0, below.

3.0 Datataking for the Hall A BCM Linearity Test

1. Restore **MBH0L02H** and **V** to the FC#2_out values recorded in [Step 5 on page 2](#) (FC#1 and FC#2 should still be in).
2. Set the beam current to the maximum test value (as determined in [Step 8 on page 3](#)).
3. Call the Hall A counting house and ask them to begin their data-logging process. After they inform you that they are logging data, wait 90 seconds and proceed to

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the following step (refer to Figure 3-1, below, for a graphical depiction of the entire data-logging process).

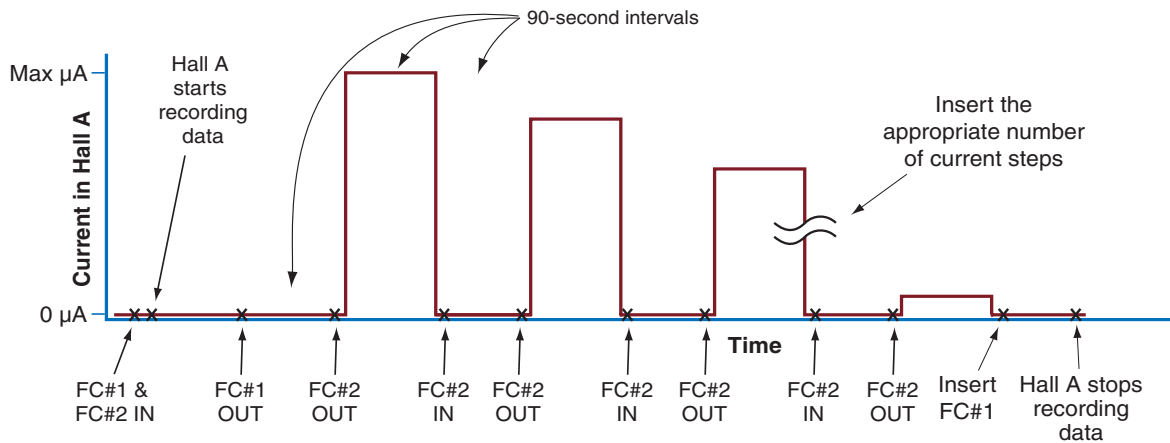


Figure 3-1: Graphical Depiction of the Data-Logging Process

4. Pull FC#1 out (leave FC#2 in), wait 90 seconds and then proceed to the following step.
5. Pull FC#2, wait for 90 seconds and then proceed to the following step.
6. Reinsert FC#2, wait for 90 seconds (while waiting, change the current to the next lowest test current) and then return to Step 5, above. Repeat this for all of the remaining test currents (refer to Figure 3-1).

NOTE: If the tune of the machine is such that injector BLM trips occur when FC#2 is inserted or removed, use FC#1 to “buffer” the insertion and removal of FC#2.

7. After all test currents are completed, insert FC#1 for 90 seconds.
8. Call the Hall A counting house and tell them to stop their data acquisition process.
9. Are you going to be performing the [Hall A Unser/OL02 Cross Calibration Procedure](#) (per a request from Hall A)?

NO **YES** →

Go to
[Step 10](#)

A. Stop the Striptool on the FC#2 picoammeter and the OL02 BCM.

B. Go to the [Hall A Unser/OL02 Cross Calibration Procedure](#), where you can skip the setup steps.

10. Resume beam deliver to the other halls scheduled to receive beam.
11. Perform the following backout steps.
 - a. Open the Current Monitoring in the Injector/Polarized Source screen (monticello.edl⇒Injector⇒INJ Current Monitoring) and reconnect FC#2 to its I to V amplifier by clicking on the I to V Amp button.
 - b. Close the Striptool that is recording the FC#2 picoammeter/OL02 BCM.s
 - c. Close all Xterm windows associated with this procedure.

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NOTE: It is a good idea to contact Hall A at this point and verify that the following conditions are what they want.

d. Turn ON the Hall A fast raster.

e. Unmask the Hall A fast raster FSD bit on FSD node 1H002.

12. PROCEDURE COMPLETE.

